

Matthew Justine M. Viray

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SUMMARY

Aspiring data professional with project-based experience and a strong foundation in data analysis and data-driven decision-making, with a broad interest across all areas of data. Familiar with Python, SQL, data visualization, statistics, and machine learning through academic and personal projects. Eager to gain real-world experience, learn from industry professionals, and contribute to impactful data work.

EDUCATION

University of California, San Diego, Current GPA: 3.85

La Jolla, CA

Bachelor of Science in Data Science

Sept 2024 – Present

College of the Canyons, Major GPA: 3.9

Santa Clarita, CA

Completed Transfer Coursework

June 2020 – June 2024

- Completed various collegiate-level courses while in high school.
- Continued taking classes after graduation to fulfill Fall 2024 transfer requirements.

PROJECTS

Airbnb Listing Popularity Analysis | *Python, Pandas, NumPy, Seaborn, Matplotlib, Scikit-learn, Tableau*

- Cleaned and transformed Airbnb data and engineered a popularity score from reviews and ratings.
- Conducted exploratory data analysis (EDA) to identify key drivers of listing performance, revealing the impact of price, location, room type, amenities, host responsiveness, and cleanliness.
- Built an interactive Tableau dashboard to visualize trends and support data-driven recommendations for improving listing visibility and guest engagement.

NYC Taxi Analysis & Predictive Modeling | *SQL, Python, Pandas, NumPy, Scikit-learn, XGBoost, Tableau, Streamlit*

- Built an end-to-end data science pipeline to clean, transform, and integrate 7M+ NYC taxi trip records using SQL and Python.
- Conducted exploratory data analysis (EDA) to identify demand, revenue, and weather-driven trends, presenting insights through interactive Tableau dashboards.
- Built a taxi demand forecasting model using cleaned NYC taxi data with location, time, and weather features, improving from a baseline regression to an XGBoost model to predict taxi counts and identify when and where there is demand.

Power Outages: Causes and Severity Analysis | *Python, Pandas, NumPy, Plotly, Scikit-learn*

- Cleaned and analyzed 1,500+ major U.S. power outage records, engineering severity metrics from outage duration and customers affected, and performing EDA across geographic, economic, and climatic factors.
- Applied statistical testing (permutation tests, KS tests, TVD) and interactive Plotly visualizations to identify seasonal, regional, and climate-driven patterns in outage causes and missingness.
- Built and tuned classification models (Decision Tree, Random Forest) to predict weather-related outages, achieving 82.1% accuracy and validating recall fairness across high- vs low-utility investment regions using permutation testing.

WORK EXPERIENCE

Hardlines Team Associate

Santa Clarita, CA

Walmart

Sept 2022 – Sept 2024

- Managed inventory and organized data-driven merchandising to optimize product availability.
- Analyzed workflow and adapted processes to improve efficiency, demonstrating problem-solving and teamwork.
- Provided customer insights by resolving inquiries and identifying patterns in store operations.

TECHNICAL & SOFT SKILLS

- **Programming:** Python, SQL, Java
- **Libraries:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn
- **Tools:** Git, GitHub, Tableau, Excel, Streamlit, VS Code
- **Soft Skills:** Communication, Teamwork, Adaptability, Problem Solving, Time Management, Attention to Detail